Jiayun Li

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At EMNLP 2026: presenting **ELITE** (Main Conference), the first framework for entity alignment on *continuously evolving* knowledge graphs, with $\sim$4$\times$ faster inference and $\sim$10$\times$ less memory than full retraining at state-of-the-art accuracy.

Looking for: research scientist, applied scientist, and postdoctoral positions after completing my PhD (expected Nov 2026).

RESEARCH INTERESTS

I study how to keep knowledge **consistent and usable when it is heterogeneous, conflicting, and changing over time**, and how to make LLM systems reason over it reliably.

- **Temporal & evolving knowledge graphs:** entity alignment across sources and time; incremental learning on dynamic graphs; principled, loss-grounded message weighting in GNNs.
- **LLMs grounded in structured knowledge:** retrieval-augmented generation, KG-grounded dialogue agents, multi-hop reasoning.
- **Alignment & evaluation:** SFT vs. DPO preference optimization; calibrated, auditable evaluation without LLM judges.

PUBLICATIONS

[1] **Jiayun Li**, et al. "ELITE: Evolving Localized and Informative Temporal Entity Alignment." **EMNLP 2026** (Main Conference). [code]

[2] **Jiayun Li**, et al. "RCTEA: Richness-guided Co-training for Temporal Entity Alignment." **Findings of ACL 2026**, pp. 39295–39310. [code]

[3] **Jiayun Li**, et al. "HTEA: Heterogeneity-aware Embedding Learning for Temporal Entity Alignment." **WSDM 2025**, pp. 982–990. [code]

[4] **Jiayun Li**, et al. "Beyond Richness: Loss-Grounded Neighbourhood Utility for Temporal Entity Alignment" (RUNE-TEA). *Under review at IEEE TKDE.*

[5] **Jiayun Li**, et al. "SPOT: Socratic Probing over Typed-graphs for Detecting Student Blind Spots." *In preparation, targeting ACL 2027.* [code]

RESEARCH HIGHLIGHTS

ELITE — Entity alignment on evolving knowledge graphs [1] 2025 – 2026

- Formalized *Evolving TEA*, replacing the static-snapshot assumption of all prior temporal entity alignment work.
- Informative seed selection plus localized incremental inference over only newly evolved entities: $\sim$4$\times$ faster, $\sim$10$\times$ less memory, SOTA accuracy.
- Rebuilt $O(N^2)$ similarity search at $\sim$180K-entity scale as chunked top-k retrieval ($>$80% memory reduction) on H100 clusters.

RCTEA & HTEA — Alignment under uneven, heterogeneous structure [2,3] 2023 – 2025

- **RCTEA:** richness-guided dual-view (structural/temporal) GAT co-training with neighborhood-consensus denoising; $\sim$6% over the strongest baselines on YAGO-WIKI180K and BETA ($p < 0.01$).
- **HTEA:** showed prior methods assume homogeneous structure and degrade on multi-source data; proposed heterogeneity-aware embedding learning, released as an open-source framework.

RUNE-TEA — From neighbour richness to loss-grounded utility [4] 2025 – 2026

- Showed that richness measures how much evidence a neighbour has, not whether it helps: on BETA, the 2.34% highest-degree senders draw 48.49% of attention, yet removing their messages *improves* H@1.
- Defined *neighbourhood utility* as the receiver- and feature-type-specific change in alignment loss when one message is removed; proved it equals the message's pointwise conditional information about the correct counterpart, and that up-weighting high-utility messages is a first-order descent step.
- RUNE-TEA learns utility from seeds and adds it as a bounded residual to richness-guided attention: outperforms all baselines on BETA and YAGO-WIKI180K, +2.35/1.09/1.91 H@1 points over RCTEA under full-candidate evaluation, at $\sim$20% extra training time.

SPOT — KG-grounded Socratic agent for knowledge blind spots [5] 2026

- Found that holistic LLM-judge scoring shows recency and fluency bias on real multi-turn transcripts; replaced it with per-concept Beta–Bernoulli beliefs over atom-level answer decomposition.
- Built a construct-validity protocol (validated student simulator, ablations, entailment-based Coverage/Detection F1) that makes verdicts auditable without ground truth.

SFT vs. DPO: when does preference optimization pay off? 2026

- Controlled LoRA study on Qwen2.5-1.5B-Instruct with judge-free metrics (behavior rate, reference perplexity, per-token KL), including 453 on-policy preference pairs.
- Off-policy pairs cost $\sim$4.2$\times$ more KL for equal suppression; SFT suppresses with N=50 where DPO needs $\sim$300; DPO's per-token mechanism shifts with training dose.

Oil & Gas Knowledge Graph + LLM QA Platform 2026

- Tiered RAG over real-world global energy asset data (dual Neo4j/PostgreSQL backends): templated retrieval, two-stage synthesis, or multi-hop retrieval chosen per query to balance cost and latency.

EDUCATION

PhD — School of EECS, The University of Queensland 2023 – Nov 2026 (expected)

Thesis: *Information Extraction from Large Knowledge Bases*. Advisors: Prof. Xue Li and Prof. Wen Hua. Australian Government RTP Scholarship.

Master of Data Science — The University of Queensland 2021 – 2022

GPA 6.19/7 (Distinction).

B.Eng., Measurement & Control Technology and Instruments — Tianjin University 2016 – 2020

EXPERIENCE

Research Officer (part-time), The University of Queensland Jul 2026 – present

Leading SPOT [5] from problem formulation to system, evaluation, and paper.

Tutor, The University of Queensland Jul 2025 – present

INFS2200 Relational Databases (2025, 2026); DATA7703 Machine Learning for Data Scientists (2026).

HONOURS & TALKS

- Australian Government Research Training Program (RTP) Scholarship (full PhD funding)
- WSDM 2025 Student Travel Grant; presented HTEA in Hannover, Germany
- Presenting ELITE at EMNLP 2026, Budapest, Hungary

SKILLS

ML/NLP: PyTorch, GNNs (GAT), Transformers, LoRA fine-tuning (SFT, DPO), RAG, agents, evaluation design
Systems: Python, Java, C/C++, SQL; Neo4j, PostgreSQL; Linux HPC and H100 clusters, mixed precision, Docker, Git
Languages: English (working language), Mandarin (native)

REFERENCES

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